# CUSTOMER SUPPORT TICKETING SYSTEM

SDLC Project Scheduling — Based on System Request & Feasibility Study

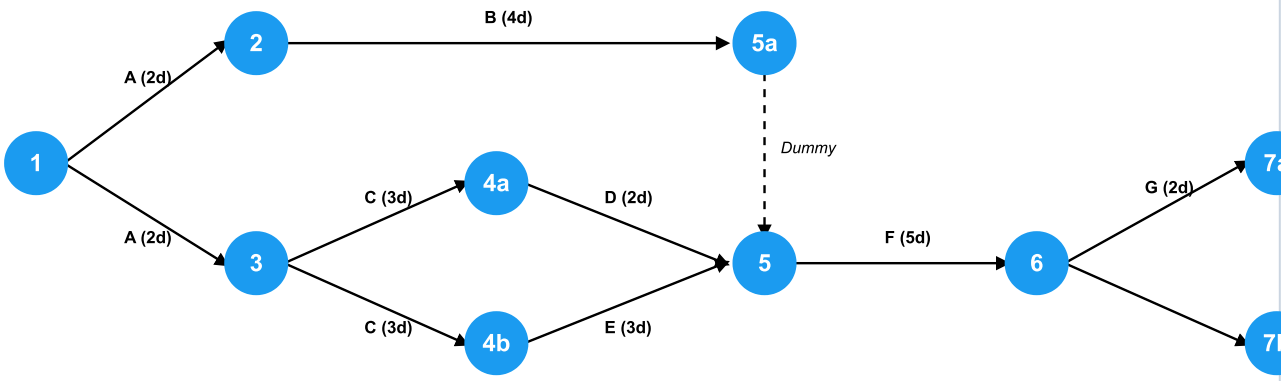
**System Activities Table**

The following table maps system development tasks based on the Customer Support Ticketing System analysis:

| Activity | Description (Derived from System Request & Feasibility) | Predecessor | Duration (Days) |
|---|---|---|---|
| A | System Request & Scope Definition | - | 2 |
| B | Feasibility Study (Economic, Technical & Operational) | A | 4 |
| C | Requirements Elicitation (Ticket Workflow & Roles) | A | 3 |
| D | Database Schema Design (MySQL / PostgreSQL) | C | 2 |
| E | UI/UX Interface Design (HTML/CSS Dashboard) | C | 3 |
| F | Core Ticketing Engine & Auto-Routing Development | B, D, E | 5 |
| G | Security, SLA Controls & Access Controls | F | 2 |
| H | System Testing, QA & User Acceptance Testing (UAT) | F, G | 4 |

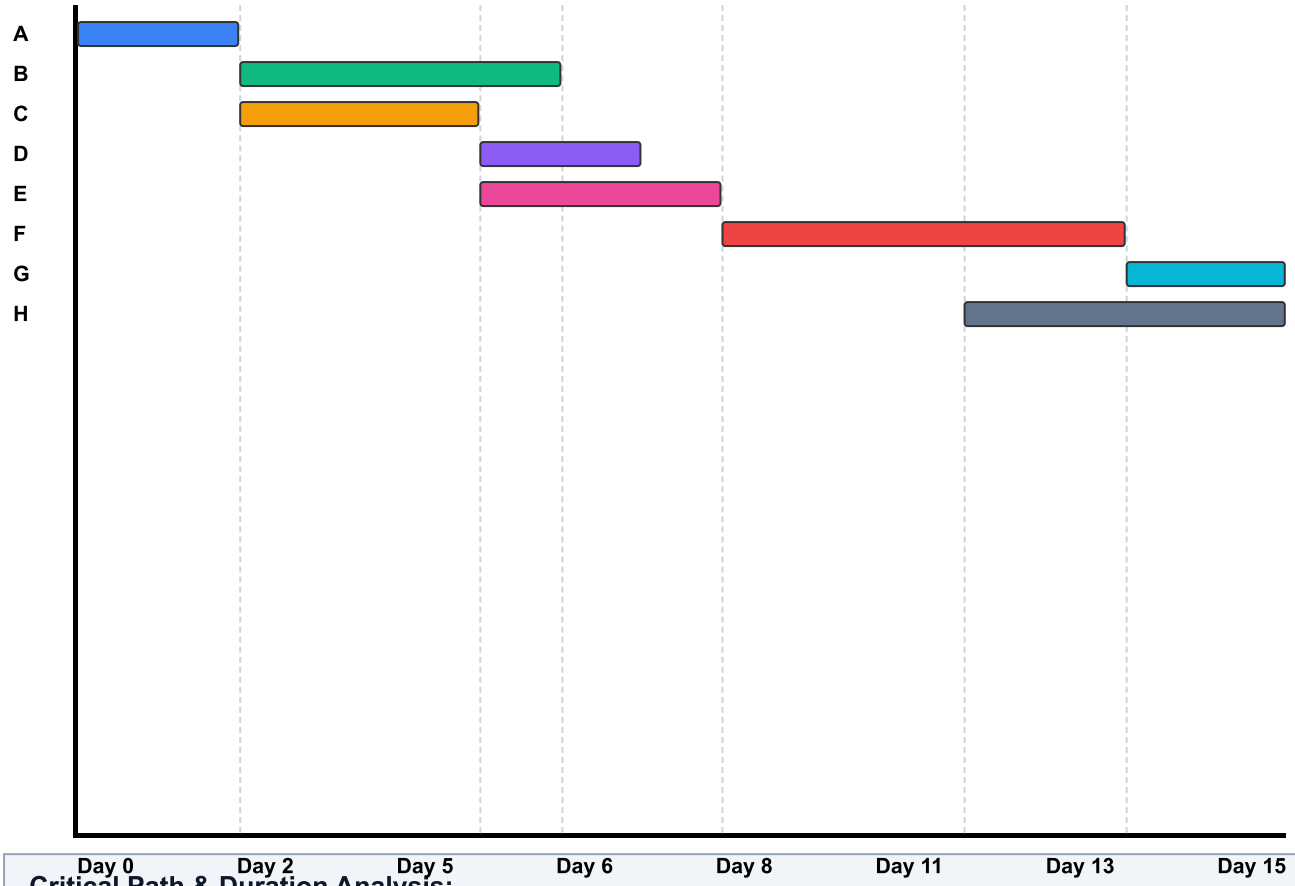
## TASK 1 & TASK 2: PERT DIAGRAM SIMULATION

Draw the PERT network diagram representing activities sequence and dependencies:



## TASK 3: GANTT CHART & MINIMUM PROJECT DURATION

> **Requirements:** Draw the Gantt Chart, calculate the Critical Path, and determine the Minimum Total Execution Time for the Ticketing System.




### Critical Path & Duration Analysis:

**Possible Paths Comparison:**

- Path 1: A ➔ B ➔ F ➔ H = 2 + 4 + 5 + 4 = 15 Days
- Path 2: A ➔ C ➔ D ➔ F ➔ H = 2 + 3 + 2 + 5 + 4 = 16 Days
- Path 3: A ➔ C ➔ E ➔ F ➔ H = 2 + 3 + 3 + 5 + 4 = **17 Days (Longest Path)**

**Critical Path:** A ➔ C ➔ E ➔ F ➔ H

**Minimum Project Completion Time: 17 Days**